

C# Decision Structures Quiz Questions

1. In C#, what statement is used to write a single-alternative decision structure?

- A. switch
- B. if
- C. while
- D. for

Answer: B

2. What appears inside the parentheses of an if statement in C#?

- A. A string literal
- B. A Boolean expression
- C. An integer value
- D. A variable declaration

Answer: B

3. In the general format of the if statement with multiple statements, what encloses the conditionally executed statements?

- A. Parentheses ()
- B. Square brackets []
- C. Curly braces {}
- D. Angle brackets <>

Answer: C

4. In the Test Score Average application, how many test scores does the user enter?

- A. Two
- B. Three
- C. Four
- D. Five

Answer: B

5. In the Test Score Average application, a congratulatory message is displayed if the average is greater than what value?

- A. 90
- B. 95
- C. 100
- D. 85

Answer: B

6. In the Test Score Average application, what happens when the Clear button is clicked?

- A. Calculates the average
- B. Clears the TextBoxes and average
- C. Exits the application
- D. Displays a message

Answer: B

7. In the code for the Test Score Average application, what constant is used for the high score value?

- A. 90.0
- B. 95.0
- C. 100.0
- D. 85.0

Answer: B

8. What is the general format of the if-else statement in C#?

- A. `if (expression) { statements; } else { statements; }`
- B. `if (expression) statements; else statements;`
- C. `switch (expression) { case: statements; }`
- D. `while (expression) { statements; }`

Answer: A

9. In the if-else statement, which part executes if the Boolean expression is false?

- A. The if clause
- B. The else clause
- C. Both clauses
- D. Neither clause

Answer: B

10. In the Payroll with Overtime application, overtime is calculated for hours worked more than what number?

- A. 30
- B. 35
- C. 40
- D. 45

Answer: C

11. In the Payroll with Overtime application, what multiplier is used for overtime pay?

- A. 1.0
- B. 1.5
- C. 2.0
- D. 0.5

Answer: B

12. In the code for Payroll with Overtime, what is the named constant for base hours?

- A. 30m
- B. 35m
- C. 40m
- D. 45m

Answer: C

13. In the Loan Qualifier application, what is the minimum salary required to qualify for the loan?

- A. \$30,000
- B. \$40,000
- C. \$50,000
- D. \$60,000

Answer: B

14. In the Loan Qualifier application, what is the minimum years on the job required?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

15. In the code for Loan Qualifier, what constant is used for minimum years on job?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

16. In the Grader application, what grading scale is used?

- A. 5-point
- B. 10-point
- C. 15-point
- D. 20-point

Answer: B

17. In the Grader application, what grade is given for scores 90 and above?

- A. A
- B. B
- C. C
- D. F

Answer: A

18. What makes the logic of nested decision structures simpler in C#?

- A. if-else-if statement
- B. switch statement
- C. while loop
- D. for loop

Answer: A

19. How many logical operators are provided in C# as shown in the table?

- A. 2
- B. 3
- C. 4
- D. 5

Answer: B

20. In the Range Checker application, what range is considered acceptable?

- A. 1 through 5
- B. 1 through 10
- C. 1 through 15
- D. 1 through 20

Answer: B

21. What data type in C# holds true or false values?

- A. int
- B. double
- C. bool
- D. string

Answer: C

22. In the example with bool grandMaster, what value is assigned if points > 5000?

- A. false
- B. true
- C. 0
- D. 1

Answer: B

23. What are bool variables commonly used as in code?

- A. Counters
- B. Flags
- C. Strings
- D. Arrays

Answer: B

24. Which operator is used to compare two strings in C#?

- A. =
- B. ==
- C. ===
- D. !=

Answer: B

25. In the Secret Word application, what is the secret word?

- A. Ariel
- B. Mark

- C. Mary
- D. Joe

Answer: A

26. In string comparison, are "Joe" and "joe" considered equal by default?

- A. Yes
- B. No
- C. Only if lowercase
- D. Only if uppercase

Answer: B

27. What method can perform a case-insensitive string comparison?

- A. String.Equals
- B. String.Compare with true
- C. String.ToLower
- D. String.Length

Answer: B

28. What do Parse methods throw when converting nonnumeric data?

- A. Warning
- B. Exception
- C. Error message
- D. Null value

Answer: B

29. What is a better practice than using try-catch for data conversion exceptions?

- A. Using loops
- B. Preventing exceptions with TryParse
- C. Ignoring exceptions
- D. Using switch

Answer: B

30. Which TryParse method converts a string to an int?

- A. double.TryParse
- B. decimal.TryParse
- C. int.TryParse
- D. bool.TryParse

Answer: C

31. What does TryParse return if conversion is successful?

- A. false
- B. true
- C. 0
- D. null

Answer: B

32. In validating multiple TextBoxes, what should the program do when invalid data is found?

- A. Ignore it
- B. Display a general error
- C. Display specific error for each TextBox
- D. Exit the program

Answer: C

33. In the Add Two Numbers Application, what happens if non-integer is entered?

- A. Adds anyway
- B. Shows error for specific TextBox
- C. Crashes
- D. Clears all

Answer: B

34. What is the process of inspecting input data for validity?

- A. Data conversion
- B. Input validation
- C. Exception handling
- D. String comparison

Answer: B

35. In input validation example for hours worked, what is the maximum valid hours per week?

- A. 168
- B. 400
- C. 40
- D. 100

Answer: A

36. In test score validation, what is the valid range?

- A. 0 through 50
- B. 0 through 100
- C. 0 through 90
- D. 0 through 200

Answer: B

37. What controls allow selecting one choice from several?

- A. TextBoxes
- B. Labels
- C. Radio buttons
- D. Check boxes

Answer: C

38. How many radio buttons in a group can be selected at a time?

- A. None
- B. One
- C. All
- D. Multiple

Answer: B

39. How are radio buttons grouped?

- A. Inside GroupBox or Panel
- B. Inside Label
- C. Inside Button
- D. Inside TextBox

Answer: A

40. What property of RadioButton determines if it is selected?

- A. Text
- B. Checked
- C. Visible
- D. Enabled

Answer: B

41. In code, how do you check if a RadioButton is selected?

- A. if (radio.Checked)
- B. if (radio.Text == true)
- C. if (radio.Visible)
- D. if (radio.Enabled)

Answer: A

42. In the Color Mixer application, what secondary color is from red and blue?

- A. Orange
- B. Green
- C. Purple
- D. Yellow

Answer: C

43. What event occurs when a RadioButton's Checked property changes?

- A. Click
- B. CheckChanged
- C. TextChanged
- D. Focus

Answer: B

44. In the Color Theme application, what happens when a radio button is selected?

- A. Form background changes immediately
- B. Message box appears
- C. Application exits
- D. Nothing

Answer: A

45. What controls allow multiple selections?

- A. Radio buttons
- B. Buttons
- C. Check boxes
- D. Labels

Answer: C

46. What property of CheckBox holds the displayed text?

- A. Checked
- B. Text

- C. Selected
- D. Value

Answer: B

47. In CheckBox, what does Checked = True mean?

- A. Unchecked
- B. Selected
- C. Disabled
- D. Hidden

Answer: B

48. In the switch statement, what is it an alternative to?

- A. if-else-if
- B. for loop
- C. while loop
- D. try-catch

Answer: A

49. What control displays a list of items?

- A. TextBox
- B. Label
- C. ListBox
- D. Button

Answer: C

50. When does a scroll bar appear in a ListBox?

- A. Always
- B. When items exceed display space
- C. Never
- D. On click

Answer: B

51. How do you add items to ListBox at design time?

- A. Code only
- B. Items property collection
- C. Text property
- D. Checked property

Answer: B

52. What property stores the selected item in ListBox?

- A. SelectedIndex
- B. SelectedItem
- C. Items.Count
- D. Text

Answer: B

53. What value is SelectedIndex if no item is selected?

- A. 0
- B. 1
- C. -1
- D. null

Answer: C

54. In Time Zone application, what is displayed when a city is selected?

- A. Time
- B. Time zone name
- C. Weather
- D. Population

Answer: B

55. How do you add an item to ListBox at runtime?

- A. Items.Add
- B. Items.Clear
- C. Items.RemoveAt
- D. Items.Count

Answer: A

56. In Number List application, what numbers are added to the ListBox?

- A. 1 to 5
- B. Shown in the image
- C. Random
- D. User input

Answer: B

57. What property reports the number of items in ListBox?

- A. Items.Add
- B. Items.Clear
- C. Items.Count
- D. Items.RemoveAt

Answer: C

58. What method erases all items in ListBox?

- A. Items.Add
- B. Items.Clear
- C. Items.Count
- D. Items.RemoveAt

Answer: B

59. What method removes an item at a specific index in ListBox?

- A. Items.Add
- B. Items.Clear
- C. Items.Count
- D. Items.RemoveAt

Answer: D

60. In the Adds, removes and clears ListBox items Application, what does the Clear button do?

- A. Adds item
- B. Removes selected
- C. Deletes all entries
- D. Exits app

Answer: C

61. In the if statement without braces, how many statements are conditionally executed?

- A. All following
- B. One
- C. None
- D. Multiple

Answer: B

62. In Test Score Average, how is the average rounded?

- A. 1 decimal place
- B. 2 decimal places
- C. No rounding
- D. To integer

Answer: B

63. In Payroll with Overtime, if hours ≤ 40 , how is gross pay calculated?

- A. With overtime
- B. hours * rate
- C. base + overtime
- D. rate * 40

Answer: B

64. In Loan Qualifier, what logical operator is used in the condition?

- A. OR
- B. NOT
- C. AND
- D. XOR

Answer: C

65. In Grader, how many nested if statements are used initially?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: D

66. What is the symbol for the AND logical operator in C#?

- A. ||
- B. &&
- C. !
- D. ^

Answer: B

67. In Range Checker, what operator checks if number ≥ 1 and ≤ 10 ?

- A. OR
- B. NOT

- C. AND
- D. None

Answer: C

68. In bool example, what does !grandMaster mean?

- A. grandMaster is true
- B. grandMaster is false
- C. Set to true
- D. Set to false

Answer: B

69. In string comparison example, what are name1 and name2?

- A. Mary and Mark
- B. Joe and joe
- C. Ariel and Mark
- D. JOE and joe

Answer: A

70. In String.Compare with true, what does it return if equal ignoring case?

- A. 1
- B. -1
- C. 0
- D. true

Answer: C

71. In TryParse, what is assigned to the target variable if conversion fails?

- A. null
- B. 0
- C. Exception
- D. Original string

Answer: B

72. In Add Two Numbers code, how many if statements check for valid input?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

73. In input validation for payroll, what if hours >168?

- A. Accept
- B. Error message
- C. Calculate overtime
- D. Set to 40

Answer: B

74. Default value of RadioButton Checked property?

- A. True
- B. False
- C. Null
- D. 0

Answer: B

75. In Color Mixer, if same color selected in both groups?

- A. Mix to secondary
- B. Set form to that color
- C. Error
- D. No change

Answer: B

76. How to create CheckedChanged event handler?

- A. Click the control
- B. Double-click the control
- C. Right-click
- D. Properties

Answer: B

77. In Check Boxes, how many can be selected?

- A. One
- B. None
- C. Any number
- D. All

Answer: C

78. In ListBox, first item index?

- A. -1
- B. 0
- C. 1
- D. null

Answer: B

79. In SelectedIndex property, last index is?

- A. n
- B. n-1
- C. 0
- D. -1

Answer: B

80. In Example 11, what button removes the selected item?

- A. Add
- B. Remove
- C. Clear
- D. Exit

Answer: B

Wish you all the best!

Fathy.